

Learning Objective 1.3: I can determine the polarization of an antenna and describe the bandwidth characteristics of common antenna types.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Identify the polarization.** For each plane wave propagating in $+\hat{\mathbf{z}}$, identify the polarization (linear, RHCP, LHCP, or elliptical) and — where linear — the tilt angle relative to $\hat{\mathbf{x}}$.

(a) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz)$

Only an $\hat{\mathbf{x}}$ component — **linear along $\hat{\mathbf{x}}$ (0°)**.

(b) $\vec{E} = \hat{\mathbf{x}}2\cos(\omega t - kz) + \hat{\mathbf{y}}2\cos(\omega t - kz)$

Equal amplitudes, in phase — **linear, tilted $+45^\circ$** .

(c) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz - 90^\circ)$

Equal amplitudes, $\delta = -90^\circ$, $+\hat{\mathbf{z}}$ propagation — **RHCP**.

(d) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz + 90^\circ)$

$\delta = +90^\circ$ — **LHCP**.

(e) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz) + \hat{\mathbf{y}}1\cos(\omega t - kz - 90^\circ)$

Unequal amplitudes, $\delta = -90^\circ$ — **right-hand elliptical**, $\text{AR} = 3$ (≈ 9.5 dB).

2. **Axial ratio.** A satellite downlink antenna is spec'd circularly polarized with $\text{AR} \leq 3 \text{ dB}$ across its operating band.

- (a) Convert the 3 dB axial ratio to a linear ratio (major/minor axis).

$$\text{AR} = 10^{3/20} \approx 1.41 \text{ (i.e. } \sqrt{2}\text{)}.$$

- (b) Write the two orthogonal linear components (up to a common scale) with the appropriate 90° phase, assuming the ellipse major axis is along $\hat{\mathbf{x}}$.

$$E_x(t) = 1 \cdot \cos(\omega t), \quad E_y(t) = 0.707 \cos(\omega t \pm 90^\circ)$$

Sign of the phase sets the RHCP vs LHCP sense.

- (c) An $\hat{\mathbf{x}}$ receiver picks off E_x ; a $\hat{\mathbf{y}}$ receiver picks off E_y . Compute the ratio of received *power* in dB. Should it match part (a)?

$$\frac{P_x}{P_y} = \left(\frac{1.0}{0.707} \right)^2 = 2.0 \longrightarrow 10 \log_{10}(2.0) \approx 3.0 \text{ dB}$$

Yes — the axial ratio in dB equals the received-power difference between the two aligned linear receivers.

3. Polarization loss factor.

- (a) A ground station transmits a *linearly* polarized signal along $\hat{\mathbf{x}}$. A satellite receiver is **RHCP** and pointed correctly. Compute the PLF in dB.

Linear $\leftrightarrow$ CP: $\text{PLF} = 0.5 = -3 \text{ dB}$.

- (b) Same ground station, but the satellite receiver is **LHCP**. Compute the PLF.

Still -3 dB — linear-to-CP does not depend on sense.

- (c) Two linear whip antennas: radio A vertical, radio B tilted 30° from vertical. Compute the PLF.

$\text{PLF} = \cos^2(30^\circ) = (0.866)^2 \approx 0.75 \longrightarrow 10\log_{10}(0.75) \approx -1.2 \text{ dB}$.

- (d) In one sentence, explain why aircraft-to-ground links almost always specify circular polarization on at least one end.

Airframe attitude (roll/pitch/yaw) sweeps a linear aircraft antenna through every tilt relative to the ground station; making one end CP caps the worst-case polarization loss at 3 dB regardless of attitude.

4. **Impedance bandwidth from a VSWR spec.** An antenna is spec'd “VSWR ≤ 2 from 2.30GHz to 2.55GHz.”

(a) What is the impedance bandwidth in MHz?

$$2.55 - 2.30 = 250\text{MHz}.$$

(b) What is the fractional bandwidth (FBW) in percent?

$$f_c = 2.425\text{GHz}, \quad \text{FBW} = \frac{250}{2425} \approx 10.3\%$$

(c) Convert VSWR = 2 to $|\Gamma|$ and to return loss in dB.

$$|\Gamma| = \frac{2-1}{2+1} = \frac{1}{3} \approx 0.333, \quad \text{RL} = -20\log_{10}(0.333) \approx 9.5 \text{ dB}$$

(d) Narrowband, broadband, or ultra-wideband? Likely family — patch, horn, or log-periodic?

FBW $\approx 10\%$ is **broadband**: a standard-gain horn or a well-designed half-wave dipole. A patch would need matching tricks; a log-periodic is overkill.

5. Fractional and ratio bandwidth.

- (a) A cellular antenna covers 698–960MHz (low) and 1.71–2.69GHz (high). Compute the FBW and RBW of each band.

$$\text{Low: } f_c = 829\text{MHz}, \quad \text{FBW} = \frac{262}{829} \approx 31.6\%, \quad \text{RBW} = \frac{960}{698} \approx 1.38:1$$

$$\text{High: } f_c = 2200\text{MHz}, \quad \text{FBW} = \frac{980}{2200} \approx 44.5\%, \quad \text{RBW} = \frac{2690}{1710} \approx 1.57:1$$

- (b) A UWB radar antenna covers 3.1–10.6GHz. Compute its RBW. Does it meet the FCC UWB threshold of $\text{RBW} \geq 2:1$?

$$\text{RBW} = \frac{10.6}{3.1} \approx 3.42:1$$

Yes — well above the 2:1 threshold.

- (c) Which antenna family would you pick for each of the three bands, with a one-sentence reason?

Low ($\approx 32\%$): broadband monopole or PIFA. High ($\approx 45\%$): chassis slot or stacked patch. UWB (3.4:1): Vivaldi (tapered slot) or log-periodic — self-scaling over a wide band without matching tricks.

6. **Chu-Harrington intuition.** A design must fit inside a sphere of radius $a = 15$ mm and operate at $f = 915$ MHz (ISM band).

(a) Compute λ , k , and the electrical size ka .

$$\lambda = \frac{c}{f} \approx 0.328\text{m}, \quad k = \frac{2\pi}{\lambda} \approx 19.2 \text{ rad/m}, \quad ka = 19.2(0.015) \approx 0.288$$

(b) Estimate the minimum Q from $Q_{\min} \approx \frac{1}{(ka)^3} + \frac{1}{ka}$.

$$Q_{\min} \approx \frac{1}{0.0239} + 3.47 \approx 45$$

(c) Estimate the maximum fractional bandwidth $\text{FBW}_{\max} \approx 1/Q_{\min}$ (for $\text{VSWR} \leq 2$).

$$\approx 1/45 \approx 2.2\% \quad (\approx 20\text{MHz at } 915 \text{ MHz}).$$

(d) The designer wants 100 MHz of usable bandwidth at 915 MHz. Possible in a 15 mm sphere?

No — 100MHz is $\text{FBW} \approx 11\%$, roughly $5\times$ the Chu-Harrington ceiling for a 15 mm sphere. It requires a larger antenna, or resistive loading / matching-network absorption that trades gain for bandwidth.

7. **Reading the datasheet.** A vendor offers a GPS patch specified as **RHCP**, $AR \leq 3$ dB , “bandwidth 4% ($VSWR \leq 2$)”. You are choosing the antenna for the other end of the link.

- (a) The datasheet defines handedness *from the receiver looking back toward the incoming wave*. Which convention is that, what is the part’s sense in **IEEE** terms, and what goes wrong if you pair it with a satellite transmitting IEEE RHCP?

That is the **optics/physics** convention, opposite to IEEE. The part is IEEE **LHCP**. Paired with an IEEE RHCP satellite it is a *sense mismatch* — ideally a null, and in practice only the few dB of rejection set by the real axial ratio (part (e)). This is a link-killer, not a few-dB penalty.

- (b) Convert $AR = 3$ dB to a linear ratio A , and give A^2 .

$$A = 10^{3/20} \approx 1.41, \quad A^2 \approx 2.0$$

- (c) A *linear* antenna receives from this patch. Using $PLF(\theta) = \frac{A^2 \cos^2 \theta + \sin^2 \theta}{A^2 + 1}$, find the best and worst PLF over all orientations θ , in dB.

$$p_{\max} = \frac{A^2}{A^2 + 1} \approx \frac{2}{3} \longrightarrow 10\log_{10}(0.667) \approx -1.8 \text{ dB}$$

$$p_{\min} = \frac{1}{A^2 + 1} \approx \frac{1}{3} \longrightarrow 10\log_{10}(0.333) \approx -4.8 \text{ dB}$$

Not the flat -3 dB of the ideal table — and you rarely control θ .

- (d) Show that the peak-to-null swing in dB equals the axial ratio in dB.

$$\frac{p_{\max}}{p_{\min}} = \frac{A^2/(A^2 + 1)}{1/(A^2 + 1)} = A^2$$

$$10\log_{10}(A^2) = 20\log_{10}(A) = AR_{\text{dB}}$$

Here $10\log_{10}(2.0) \approx 3.0$ dB , matching the spec. The result is exact, not a rule of thumb.

- (e) The far end is an opposite-sense CP antenna, also $AR = 3$ dB , major axes aligned. Worst-case rejection is $\left(\frac{A^2 - 1}{A^2 + 1}\right)^2$. Evaluate it and compare with the idealized PLF table.

$$\left(\frac{2 - 1}{2 + 1}\right)^2 = \frac{1}{9} \approx 0.111 \longrightarrow 10\log_{10}(0.111) \approx -9.5 \text{ dB}$$

(Carrying $A^2 = 1.995$ instead of 2 gives -9.6 dB .) The ideal table says $-\infty$ dB . Real hardware gives roughly 10 dB of cross-pol isolation — useful, but nowhere near a null.

- (f) Name two things you must ask before believing the antenna works over the advertised 4%.

Which bandwidth — 4% is the *impedance* BW at $VSWR \leq 2$, while the axial-ratio and pattern bandwidths are narrower and go unquoted — and **over what scan angle**, since AR is normally specified at boresight only and degrades off-axis. A defensible third: at what VSWR threshold, since the number moves with the bar.

Documentation: _____